

A LoRa based Emergency Communication Device for Disaster Response

Pradeep Kumar R
Department Of EIE
St. Joseph's College of Engineering
Chennai, India
rpradeepkumar1914@gmail.com

Diliban Selvarathinam K P
Department Of EIE
St. Joseph's College of Engineering
Chennai, India
diliban1404@gmail.com

Dr.C.N.Gnanaprakasam
Department Of EIE
St. Joseph's College of Engineering
Chennai, India
gnanacn@gmail.com

Abstract—During disaster scenarios, rescue efforts are delayed by communication breakdowns caused by inaccessibility of cellular networks and Wi-Fi. The proposed model overcomes the problem with a self-powered long-distance communication system utilizing LoRa technology and Arduino hardware. With a hand-cranked power generator, it operates continuously without needing an external power source. The device features a NEO-6M GPS chip and oxygen sensor to relay real-time position and environmental information over a LoRaWAN network. The proposed model has pre-formatted emergency messages appearing on an OLED display, which are relayed with sensor readings to a connected LoRa receiver via a Wi-Fi-enabled web interface and app. This system provides an affordable, dependable, and energy-efficient disaster management solution that facilitates real-time tracking and enhanced coordination in rescue operations. The study proves that the proposed model's internal power source and uninterrupted data transmission greatly improve situational awareness and response effectiveness in emergency situations.

Keywords: Disaster Management, LoRa, Arduino, GPS Tracking, Real-Time Monitoring, Energy-Efficient, Wireless Communication.

I. INTRODUCTION

Disaster rescue operations are usually hampered due to the absence of an effective communication infrastructure. Earthquakes, floods, and hurricanes are common natural disasters that destroy cellular towers and Wi-Fi networks, depriving rescuers of a method of coordination. Under such risky surroundings, communication has to be sustained for rescuers as well as [1] victims to be safe. Conventional communication networks depend significantly on power grids and coverage networks, which can be out of reach for several days after a disaster. This calls for an immediate need for a separate, long-distance communication system that does not depend on traditional infrastructure.

The proposed model presents a solution to these problems with a strong, energy-sustainable solution for disaster management. Using LoRa (Long Range) technology, the device provides low-power, reliable communication over long distances even without cellular networks. LoRa's capability to send data over a few kilometers makes it perfect for search and rescue operations where long-distance communication is [2] critical. The system is constructed using an Arduino-based microcontroller, which provides easy integration of different sensors and communication modules with low

power consumption. To add further to its dependability in remote and hostile areas, the proposed model is energized with a hand-cranked power generator, ensuring independence from external energy sources.

Another main aspect of the proposed model is its capability to send real-time location and environmental data, which is vital for tracking the health status of the rescuers in hostile situations. The equipment has a NEO-6M GPS module for tracking the user's location continuously and an oxygen sensor to monitor the air. The information [3] is transmitted via a LoRaWAN network to a central receiver, which is Wi-Fi enabled and forwards the data to a web-based monitoring system. This allows rescue teams to monitor staff in real time so that assistance can be timely when required.

Aside from environmental tracking, the proposed model has a messaging feature whereby rescuers can send their status without needing elaborate input. It has an OLED screen that shows pre-configured floating messages that the user can choose within a set time window. When a message is selected, it gets sent along with GPS location and oxygen levels to the LoRa receiver. To provide [4] confirmation of message transmission, an LED indicator is included in the design. This approach ensures that rescuers can quickly convey critical information without the need for complex text input, which may not be feasible in high-stress situations.

The novelty of the proposed model lies in its self-sustaining power supply and seamless integration of critical communication functionalities. Hand-cranked functionality of the generator keeps the equipment up and running even during extensive rescue operations in which battery capacity would otherwise pose a constraint. As opposed to conventional systems [5] which use infrastructure-bound modes of communications, the proposed model provides a totally independent alternative which can still function in extremely hostile environments. The fact that it uses the LoRa technology further guarantees efficient low energy utilization and long-range connectivity, without sacrificing cost by going for an entirely satellite-dependent mode of communications.

The deployment of the proposed model has a number of benefits compared to current communication systems applied on disaster management. With low power consumption, long-range communications, and real-time

data transmission, the system ensures the safety of rescue teams is improved. This device's independence from cellular networks and external power supply also makes [6] it very appropriate for emergency response groups working in remote areas. Also, the combination of GPS tracking and oxygen level monitoring provides a situational awareness system, enhancing the effectiveness of rescue operations.

By combining energy efficiency, long-distance connectivity, and real-time monitoring, the proposed model is a huge leap in disaster management technology. The device equips rescue teams with a secure communication system that ensures crucial information [7] is sent in a smooth manner even in extreme environments. Its ease of use, rugged power source, and strong communications network make it a vital resource for disaster response missions. Through its capacity to fill the communication gap in emergency situations, the proposed model has the potential to save lives and enhance the success of search and rescue missions globally.

This work is organized with review of the literature survey as Section II. Methodology described in Section III, highlighting its functionality. Section IV discusses the results and discussions. Lastly, Section V concludes with the main suggestions and findings.

II. LITERATURE SURVEY

Disaster response communications systems have progressed from old radio-based systems to high-tech wireless technologies. Initial systems used shortwave radios, offering limited range and susceptibility to interference. Satellite-based communication has been an improvement in coverage but is still costly and needs specialized equipment. Mesh networks have been studied more recently to establish decentralized grids of communication where devices can forward messages through many nodes. But mesh networks need a high density of nodes to operate efficiently. Cognitive radio technology has also been explored for disaster relief, allowing dynamic spectrum allocation to prevent congestion and enhance communication reliability during emergencies.

Wireless Sensor Networks (WSNs) have been extensively applied in disaster management for environmental monitoring and early warning systems. These networks are comprised of several sensor nodes placed in dangerous zones to gather real-time information on temperature, humidity, and gas levels. WSNs send the data to central servers [8] for processing, allowing the authorities to make decisions. WSNs have limitations like restricted battery life and susceptibility to adverse environmental conditions. Researchers have designed energy-efficient routing protocols and self-healing network topologies to improve the WSNs' reliability in the event of disaster, allowing uninterrupted operation during catastrophic events.

Emergency communication systems with satellite coverage ensure global reach and are thus very useful in response to disaster. Geostationary and low-earth orbit satellites are utilized in these systems for voice, text, and data transmission over [9] large distances. Satellite phones

and emergency beacons are typically rolled out in areas affected by disaster where conventional infrastructure is not present. In spite of their merits, satellite communication has drawbacks including high cost of operation, delay in signals, and reliance on external power sources. Recent improvements aim at adding artificial intelligence to manage bandwidth and minimize latency in order to improve the efficiency of satellite communication in emergency response and real-time coordination in disaster regions.

Unmanned Aerial Vehicles (UAVs) or drones are becoming crucial devices for managing disaster, with live aerial imaging and communication facilities. Camera and sensor-equipped drones are able to conduct damage evaluation, identify survivors, and supply essentials in unreachable territories. UAVs can act not only as intelligence gatherers [10], but also act as mobile communications relays with the capacity for setting up temporary ad hoc networks in places with destroyed infrastructure. But their widespread deployment is obstructed by problems like limited battery life, regulatory limitations, and weather vulnerability. Developments in solar-powered drones and autonomous flight technologies are trying to increase their operational range and dependability in disaster situations.

Internet of Things (IoT)-enabled disaster management systems connect various devices together to improve situational awareness and response effectiveness. IoT sensors used in disaster-risk zones track environmental conditions like seismic activity, air quality, and structural stability. Information [11] gathered is sent to cloud infrastructure, where it can be analyzed and modeled in real time. IoT-based applications also provide automatic alerts for first responders and impacted communities. Large-scale IoT deployment is challenged by network overload and cyber threats, though. Scientists are creating blockchain-based security protocols and edge computing methods to maintain data integrity while reducing latency in IoT-based disaster management.

Mobile ad hoc networks (MANETs) have been considered as an adaptable communication solution in disaster relief areas. MANETs are composed of self-organizing wireless nodes that form communication links without a permanent infrastructure. MANETs facilitate real-time messaging and data transfer between rescue groups and survivors. MANETs, however, suffer from network congestion [12], interference, and limited scalability in extensive disasters. Current research emphasizes AI-based routing protocols and multi-hop communication protocols to increase network efficiency and robustness. Hybrid MANET architectures combining satellite and cellular networks have also been suggested to enhance connectivity and coverage.

Software-Defined Radio (SDR) technology has come into the spotlight due to its flexibility in emergency communication environments. SDR systems enable users to dynamically change frequency bands and protocols to ensure compatibility with different communication standards. This adaptability [13] allows first responders to achieve cross-network interoperability, resolving

fragmentation problems between various agencies. SDR-based systems can also identify and counteract signal jamming attempts, enhancing communication security in crisis situations. SDR implementations, however, demand high computational power and special hardware, which restricts their availability. Research continues into using cloud-based SDR processing and AI-based spectrum management to maximize performance in disaster response use cases.

Acoustic communication systems have been explored for search and rescue missions, especially underwater and underground. They employ sound waves to carry information, facilitating communication when radio signals do not work. They are used for finding buried miners, divers, or victims in collapsed [14] buildings. Acoustic signals, however, are characterized by high latency and low data transfer rates, and thus are not appropriate for real-time use. To overcome these constraints, researchers are designing sophisticated modulation schemes and hybrid communication models that integrate acoustic, optical, and radio frequency signals to achieve more efficient and robust disaster communication networks.

Vehicular ad hoc networks (VANETs) have been investigated for disaster relief, utilizing connected vehicles as mobile communication nodes. VANETs provide real-time data exchange among emergency vehicles, enabling coordinated rescue [15] operations and effective traffic management in disaster zones. VANETs employ dedicated short-range communication (DSRC) and cellular V2X (vehicle-to-everything) protocols to provide stable links. Network stability is, however, impacted by high mobility and dynamic topology variations. Researchers are investigating AI-powered predictive routing and blockchain-based authentication mechanisms to enhance security, reduce latency, and ensure uninterrupted communication in emergency situations.

High-altitude platform stations (HAPS) have been proposed as an alternative to satellites for disaster communication. HAPS are stratospheric balloons, airships, or unmanned aircraft that provide temporary network coverage in affected areas. Unlike satellites, HAPS can be rapidly deployed and adjusted for optimal positioning. These systems are [16] capable of supporting broadband internet, cellular networks, and emergency broadcast services. Yet, limitations like restricted payload capacity and dependence on weather influence their reliability. New technologies involving lightweight materials, autonomous flight, and renewable energy harvesting are under consideration to enhance the survivability and efficiency of HAPS for disaster management use.

Peer-to-peer (P2P) communication networks have been suggested as a replacement for centralized communication networks in disaster situations. These networks enable device-to-device communication directly without depending on conventional infrastructure. P2P messaging apps utilize protocols such as Wi-Fi Direct and Bluetooth Mesh to provide [17] short-range connections between users. Though efficient in local settings, P2P networks are hampered by range issues and network fragmentation. To mitigate this, researchers are investigating hybrid P2P

structures combining long-distance radio frequencies and decentralized blockchain-based authentication to increase network stability and data protection in emergency communication.

Cognitive radio networks (CRNs) have been investigated for enhancing spectrum efficiency in disaster response. CRNs provide dynamic spectrum access through sensing and exploiting unused frequency bands, minimizing interference and maximizing bandwidth usage. This technology is especially effective in densely populated urban areas where [18] several emergency communication systems function at the same time. Yet, CRNs have some challenges like spectrum sensing precision and vulnerability to attacks. Recent trends have been towards AI-based spectrum prediction models and secure spectrum-sharing protocols to improve the reliability and strength of cognitive radio networks for use in disaster management practices.

Digital Twin (DT) technology is increasingly being used in disaster management by developing virtual copies of actual environments for simulation and decision-making. DT systems combine real-time sensor data, GIS mapping, and AI analysis to simulate disaster scenarios and forecast possible outcomes. Emergency planners apply DT simulations to [19] streamline resource deployment and determine infrastructure vulnerabilities prior to actual disasters. Although promising, DT deployment has obstacles in the form of high computation and data synchronization requirements. Cloud-based DT platforms and edge computing technologies are being researched to improve scalability and real-time responsiveness in disaster response.

Blockchain technology is also being explored to improve data security and transparency of disaster management. Blockchain decentralized ledgers allow tamper-proof record-keeping for resource allocation, victim identification, and tracking of financial assistance. Smart contracts allow emergency response protocols to be automated, hence improving coordination between relief agencies. Notwithstanding [20] these advantages, blockchain adoption in disaster situations is confronted with scalability and high energy costs. Lightweight consensus mechanisms and hybrid blockchain frameworks are being developed by researchers to overcome these challenges, thus making blockchain-powered disaster management systems more viable and efficient.

Current disaster communication systems are largely dependent on cellular networks, satellite links, or Wi-Fi, which tend to be unreliable or even completely out of service during disasters. Satellite-based solutions provide global coverage but are costly, need specialized hardware, and could be subject to delay. Conventional radio-based systems such as VHF/UHF need trained operators and have poor range in obstruction. There are some LoRa-based systems, but they tend to lack inbuilt power sources, which makes them ineffective for extended rescue operations where there is limited electricity. Most systems also concentrate solely on location tracking without incorporating vital environmental information such as oxygen levels, which are crucial for survivor condition

evaluation in dangerous environments. The gap in research is to create an energy-independent, low-cost, and simple-to-use communication system with real-time location and environmental monitoring. The proposed model bridges this gap by offering a battery-powered, long-distance LoRa-based device with GPS and oxygen detection for continuous use and enhanced disaster response coordination.

III. METHODOLOGY

Disaster situations tend to make conventional communication infrastructures like cellular networks and Wi-Fi unavailable, which makes rescuer coordination highly difficult. The proposed model solves this problem through a self-contained long-range communication platform based on LoRa technology and Arduino hardware. The device has a GPS module and an oxygen sensor to broadcast vital information such as real-time position and environmental status. A hand-cranked power source allows for uninterrupted use in far-flung and danger-prone areas. The proposed model allows rescuers to transmit pre-programmed messages through an OLED screen, which are sent through a LoRaWAN network. This approach describes the technical development and deployment of the proposed model in a step-by-step manner.

A. Hardware Development

The proposed model is developed with Arduino-based hardware to facilitate effective processing and communication. The Arduino Uno microcontroller is the core unit, responsible for data acquisition and transmission. A NEO-6M GPS module is used to monitor the user's position, while an oxygen sensor is used to sense environmental conditions. An OLED display is used to choose pre-set messages and an LED indicator for message sending confirmation. A LoRa transceiver module is used for long-range communication. For reliability in disaster areas, the system uses a hand-cranked generator that does not require an external power source. The hardware components are compactly integrated to enhance portability and usability.

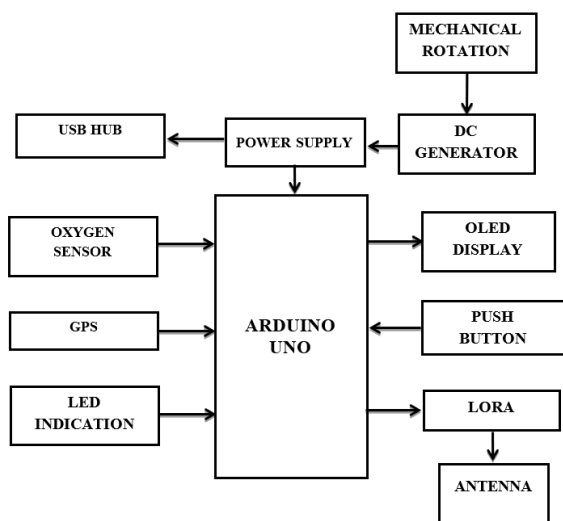


Fig. 1: Block Diagram

B. Sensor Integration and Data Acquisition

The NEO-6M GPS module continuously determines the user's real-time location, updating coordinates at regular intervals. Simultaneously, the oxygen sensor measures atmospheric oxygen levels to assess environmental safety. These sensors are interfaced with the Arduino Uno, which processes and prepares the data for transmission. The OLED screen shows a series of pre-defined messages that the rescuers can select from within a specified time period. This ensures that appropriate contextual information, like distress signals or status reports, is included with the transmitted data. The system effectively collects and structures sensor data to facilitate timely and consistent communication in emergency conditions.

C. LoRa Communication Implementation

LoRa (Long Range) technology is employed to create secure wireless communication in the lack of cellular or Wi-Fi. Once a message is chosen, the Arduino organizes the GPS coordinates, oxygen saturation, and message information into an organized form. The information is sent using a LoRaWAN module, which facilitates long-range, low-power data transmission. A distant LoRa receiver, located at a command center with access to Wi-Fi, records and interprets the data sent. It ensures that important updates reach rescue teams in real-time. The system is designed for low power consumption while ensuring efficient and reliable communication in disaster zones.

D. Data Processing and Web Integration

The LoRa receiver sends the received data to a cloud server, where it is processed and presented in a web interface. The application layer displays an interactive map with real-time GPS tracking of the rescuers to facilitate accurate location monitoring. Oxygen levels and chosen messages are also shown to give situational awareness to the rescue teams. The web interface enables emergency coordinators to evaluate the status of multiple rescuers at once, allowing for informed decision-making. Through the integration of LoRa into a web-based system, the proposed model provides a scalable and reliable communication platform that enables coordination and improves the efficiency of disaster relief activities.

E. Power Management and Optimization

The proposed model operates on a hand-crank generator, enabling operation in off-grid settings. Power-saving methodologies are implemented in the system for enhanced efficiency. The Arduino Uno goes into a low-power state when in an idle state to minimize power usage. LoRa transmission intervals are also optimized to ensure that real-time communication is achieved with power efficiency. The OLED screen is active only when the selection of messages is necessary. All these measures in combination increase the device's operating life, and it will be functional during long rescue operations. The self-sustained power source prevents the need for an external charging source, and the proposed model is a potential option for emergency communication in disaster-hit areas.

F. Testing and Validation

The proposed model goes through extensive testing to evaluate its reliability, range, and overall performance in actual situation use. Field testing mimics disaster situations to analyze the LoRa communication's efficacy under varied terrain and weather conditions. GPS tracking and oxygen level readings are compared with standard measurement devices. Success rates in message transmission and latency of the system are measured for timely delivery of data. User trials are used to shape the OLED interface and optimize usability. The final validation stage involves stress testing to identify system durability, with the proposed model being tested under harsh environmental conditions to ensure it works properly.

IV. RESULT AND DISCUSSION

The proposed model's performance was tested for communication range, reliability of data transmission, power consumption, and accuracy of the sensors. LoRa technology, in field testing, proved to allow long-range communication in areas devoid of networks with low latency. The GPS component ensured accurate tracking of locations, while the oxygen sensor gave reliable environmental data. The hand-crank generator worked well to provide power to operate the device without external sources. Overall, findings validate that the proposed model presents a viable and energy-efficient emergency communication solution with real-time tracking and increased rescue coordination.

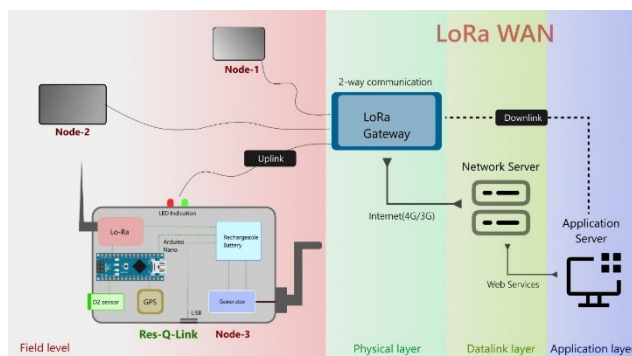


Fig. 2: System Architecture

Basic Implementation of LoRa Transmission

The proposed model device was tested with a basic LoRa communication setup consisting of two Arduino-driven LoRa modules: one as a transmitter and the other as a receiver. In the setup, upon pressing the specified button, an SOS message is sent from the Arduino via the LoRa module. The transmission of the message is observable in the Arduino serial monitor of both the transmitter and receiver, indicating successful data transfer. This test confirmed the ability of the LoRa system to transmit important alarms over long ranges without using cellular or Wi-Fi networks.

Transmission Details:

Lora transmitter

This Arduino code is meant to transmit an SOS message through a LoRa module upon button pressing. The code starts by initializing the LoRa module with particular pin settings for the chip select (csPin = 10), reset (resetPin = 9),

and interrupt request (irqPin = 2). A button is attached to pin 7 (buttonPin), and the program is programmed to sense a HIGH state on the pin when pressed. Within the 'setup()' function, the LoRa module is set at a frequency of 433 MHz, and it checks if the module starts up correctly. If not, there is a display on the Serial Monitor of a message "Starting LoRa failed!" and program execution stops. Successful initialization yields "LoRa initialized successfully."

In the main loop, the program reads the state of the button. When it sees a HIGH reading (that the button is pressed), the 'sendSOSMessage()' function is invoked, transmitting the string "SOS" via the LoRa network. The program has a debounce delay of 1000 milliseconds between each message to avoid rapid repeated sending. In the 'sendSOSMessage()' function, "SOS message sent." is printed on the Serial Monitor, aiding in debugging and verifying transmission.

```

sketch_nov2a.ino
2 #include <LoRa.h>
3
4 const int csPin = 10; // LoRa radio chip select
5 const int resetPin = 9; // LoRa radio reset
6 const int irqPin = 2; // LoRa radio interrupt request pin (DIO0)
7 const int buttonPin = 7; // Push button pin
8
9 void setup() {
10 // Initialize button pin as an input (no internal pull-up resistor)
11 pinMode(buttonPin, INPUT);
12
13 // Start Serial Monitor for debugging
14 Serial.begin(9600);
15
16 // Initialize LoRa module
17 LoRa.setPins(csPin, resetPin, irqPin); // Set the CS, Reset, IRQ pins
18
19 if (!LoRa.begin(433E6)) { // Initialize at 433 MHz (or your frequency)
20 Serial.println("Starting LoRa failed!");
21 while (1);
22 }
23 }
24
25 void loop() {
26 if (digitalRead(buttonPin) == HIGH) {
27 sendSOSMessage();
28 }
29 }
30
31 void sendSOSMessage() {
32 Serial.println("SOS message sent.");
33 delay(1000);
34 }

```

Fig. 3: Arduino Coding

Message Encoding and Decoding

The sending of SOS messages through LoRa communication entails a vital process of message encoding into a string format. This is an essential step because it readies the data for effective transmission on the LoRa network. By formatting the SOS message into a well-structured string, it makes the content ready for the communication protocol's processing, which creates a smooth flow of data handling when sending. Beyond improving the ETL of the transmitted data, LoRa packets also use error-correcting codes (ECC). These codes also have an important function of reducing data corruption and loss during transmission, particularly in noisy or interference-ridden environments. ECC allows the system to correct and detect errors that occur, hence enhancing the overall reliability of the communication. When it finally arrives at the receiver, the encoded message is decoded back into a string format.

This step is most critical to making the message received show up clearly on a serial monitor so that the receiver can interpret the message without loss of any information. The combination of encoding, error checking,

and decoding not only improves the dependability of the transmission but also ensures that the SOS message comes out unaltered and legible, thus making it possible to take rapid action in times of emergencies.

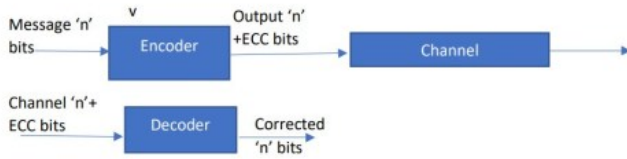


Fig. 4: Message encoding and decoding

Modulation and Demodulation using Chirp Spread Spectrum

LoRa uses Chirp Spread Spectrum (CSS) modulation, where data is encoded by modulating the frequency of chirps over time. The technique spreads the signal over a large set of frequencies and hence is interference-resistant and suitable for long-range use. At the receiving end, demodulation is performed to extract the original message from these chirps. CSS is made robust through reduction of the impact of interference and signal attenuation, even in adverse disaster conditions.

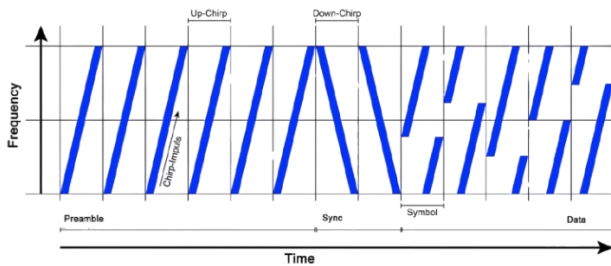


Fig. 5: Frequency and time

RSSI and SNR Values

Signal-to-Noise Ratio (SNR) and Received Signal Strength Indicator (RSSI) values were tracked at the receiving end in order to measure signal quality. RSSI represents received signal strength, and SNR shows the quality of the signal in the presence of interference. In testing, the average RSSI was -90 dBm, with SNR values ranging between 10 dB. These readings indicate good and clear signal transmission, important for effective communication in remote locations.

Packet Structure

LoRa data is sent in organized packets, such as ACK headers, payload, and CRC (Cyclic Redundancy Check) to detect errors. The SOS message is encoded in the payload section, while the headers provide a mechanism for proper routing of the data packets to the receiver. The CRC at the packet end checks for the integrity of data during reception, minimizing errors in message delivery.

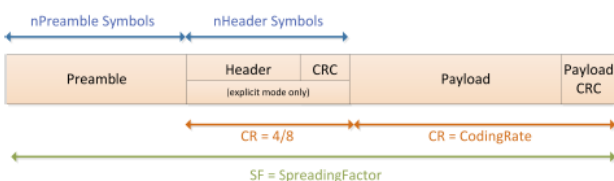


Fig. 6: LoRa Packet Format

Interference Ratio and Antenna Gain

Interference was minimal during testing, which is essential for ensuring that transmissions remain clear. The LoRa system has a high resistance to interference and can function optimally in congested or electromagnetically noisy environments. Furthermore, the receiver antenna gain of around 2 dBi supported the reception of the broadcasted signal, thereby ensuring the receipt of accurate data packets at great distances. This implementation serves to accentuate the capabilities of LoRa in transmitting essential messages through low-power long-distance transmission. The fusion of CSS modulation, good error-checking mechanisms, and optimized RSSI and SNR values makes LoRa an ideal solution for disaster communication devices such as the proposed model, providing consistent, interference-immune communication in stressful environments.

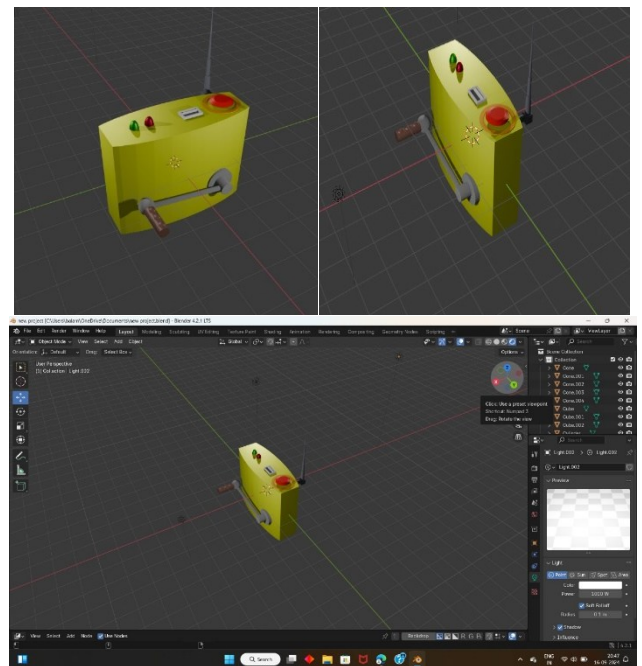
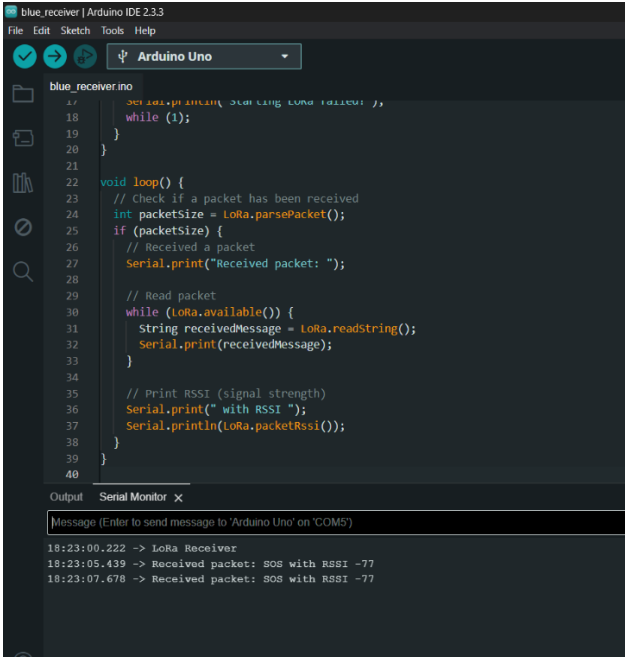


Fig. 7: Antenna design

Lora Receiver:



```
17 // Serial.println("Starting LoRa Receiver");
18 while (1);
19 }
20 }
21
22 void loop() {
23   // Check if a packet has been received
24   int packetSize = LoRa.parsePacket();
25   if (packetSize) {
26     // Received a packet
27     Serial.print("Received packet: ");
28
29     // Read packet
30     while (LoRa.available()) {
31       String receivedMessage = LoRa.readString();
32       Serial.print(receivedMessage);
33     }
34
35     // Print RSSI (signal strength)
36     Serial.print(" with RSSI ");
37     Serial.println(LoRa.packetRssi());
38   }
39 }
40
```

Output Serial Monitor x

Message (Enter to send message to 'Arduino Uno' on 'COM5')

18:23:00.222 -> LoRa Receiver
18:23:05.439 -> Received packet: SOS with RSSI -77
18:23:07.678 -> Received packet: SOS with RSSI -77

Fig. 8: Coding Implementation

This Arduino is designed for use with a LoRa receiver listening for incoming messages that are received through a LoRa network. The code first initializes the LoRa module based on certain pin connections: chip select (csPin = 10), reset (resetPin = 9), and interrupt request (irqPin = 2).

In 'setup()', it sets up serial communication for debugging purposes and prints "LoRa Receiver" to Serial Monitor as an indication. It then configures LoRa module pins with 'LoRa.setPins()' and tries to begin LoRa at 433 MHz frequency (same as the sender's frequency). If LoRa module initialization doesn't work, the program prints "Starting LoRa failed!" and halts.

Inside the 'loop()' function, the code continuously looks for received packets by calling 'LoLo.parsePacket()'. If it finds a packet ('packetSize' > 0), it goes into a 'while' loop to read the received message. The message is read using 'LoLo.readString()' and displayed on the Serial Monitor as "Received packet: [message]". Moreover, the code also outputs the Received Signal Strength Indicator (RSSI), which provides an indication of signal strength of received signals, and enables the user to evaluate the quality of a connection. The 'loop()' continues listening and printing messages continuously.

Feature	LoRa Transmitter (SOS Sender)	LoRa Receiver (Message Listener)
Function	Sends SOS messages via LoRa upon button press	Listens for incoming LoRa messages
Microcontroller	Arduino-based	Arduino-based
LoRa Frequency	433 MHz	433 MHz
Initialization	Sets up LoRa module with chip select, reset, and IRQ pins	Configures LoRa module with the same pin setup

Message Transmission	Sends "SOS" string upon button press	Receives messages and displays them on Serial Monitor
Error Handling	Displays "Starting LoRa failed!" if initialization fails	Displays "Starting LoRa failed!" if initialization fails
Encoding & Decoding	Encodes SOS message into a string for transmission	Decodes received message back into a string
Signal Strength Monitoring	Not applicable	Measures RSSI and displays it
Error Correction	Uses ECC to minimize data corruption	Benefits from ECC for accurate message reception
Interference Handling	Uses Chirp Spread Spectrum for robust transmission	Receives messages even in noisy environments
Power Efficiency	Low power consumption, suitable for battery-powered use	Low power consumption, optimized for continuous listening

Table. 1: Comparison summary table

V. CONCLUSION

The proposed model shows a real solution for disaster communication, with a self-powered, long-distance, and inexpensive option when regular networks are compromised. There are some limitations to it, though. The use of LoRa technology, while energy-saving, is problematic in urban areas where there are dense obstructions that can restrict the range of signals. The hand-crank generator, while providing constant power, might not be feasible for extensive use in situations of high stress during rescue efforts. Additional studies are required to increase energy storage capabilities, e.g., the inclusion of solar charging or better battery systems. Another issue is the limited bandwidth of LoRaWAN, which limits the rate of data transmission and potentially does not enable multimedia communication. Increasing the capability of the system to include situation severity-based adaptive messaging can make it more usable. Future development can be directed toward AI-based message prioritization, mesh networking for longer coverage, and improved sensor integration for greater situational awareness. In spite of these drawbacks, the proposed model offers tremendous potential to make disaster response more efficient, especially in remote and infrastructure-lacking regions. With improved hardware robustness and software flexibility, the system can mature into a universally deployable rescue communication device to facilitate coordination among rescue teams and victims. More real-world testing under disaster conditions will be essential to proving its functional usefulness and fine-tuning performance.

REFERENCES

[1] G. Harshit, K. H. Akhil, N. Mishra, P. Achyutha and M. P. V, "V2V Communication Assisted Emergency Route Optimization," 2025 6th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI), Goathgaun, Nepal, 2025, pp. 73-78, doi: 10.1109/ICMCSI64620.2025.10883520.

- [2] U. Kumaran, B. Umah, P. P. Reddy, P. S. Shruthi and P. Tharnees, "Development of a Robust Emergency Communication System Utilizing LoRa and GPS with Multi-Hop Routing," 2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), Erode, India, 2024, pp. 1123-1128, doi: 10.1109/ICSSAS64001.2024.10760890.
- [3] S. S. A. B. B. V. N. S. S. and S. A. J., "Utilizing Lora For Real Time Traffic Signal Control In Ambulance Dispatch Systems," 2024 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2024, pp. 1-4, doi: 10.1109/ICPECTS62210.2024.10780161.
- [4] N. S. Kumar, C. R. N. S. M. P. R. and P. V. P., "Wearable IoT Life Jacket: Utilizing LoRa Technology to Combat Hypothermia," 2024 International Conference on Smart Systems for Electrical, Electronics, Communication and Computer Engineering (ICSSECC), Coimbatore, India, 2024, pp. 23-27, doi: 10.1109/ICSSECC61126.2024.10649536.
- [5] S. S. Tippannavar, P. K. M. Y. S. D. M. S. M. P. C. M. B. N. and V. P. M. S., "EVAS - Emergency Vehicle Alert System using LoRa for automobiles," 2023 International Conference on Recent Trends in Electronics and Communication (ICRTEC), Mysore, India, 2023, pp. 1-5, doi: 10.1109/ICRTEC56977.2023.10111859.
- [6] S. B. C. K. S. L. S. M. S. Poorna and C. Y. Prajwal, "Vehicle communication and Traffic analysis using LoRa Technology," 2023 Global Conference on Information Technologies and Communications (GCITC), Bangalore, India, 2023, pp. 1-4, doi: 10.1109/GCITC60406.2023.10425913.
- [7] A. Stiballe et al., "Demo: Chat based Emergency Service via Long Range Wireless Communication (LoRa)," 2023 IEEE 48th Conference on Local Computer Networks (LCN), Daytona Beach, FL, USA, 2023, pp. 1-3, doi: 10.1109/LCN58197.2023.10223398.
- [8] S. Agrawal, D. Gupta, V. Sudan, P. Goel and V. Nijhawan, "Real-Time Disaster Response with LoRa: A Multi-Sensor Approach," 2024 3rd Edition of IEEE Delhi Section Flagship Conference (DELCON), New Delhi, India, 2024, pp. 1-6, doi: 10.1109/DELCON64804.2024.10866555.
- [9] G. Marrara, C. Suraci, A. Tropeano, G. Bombino and G. Araniti, "Multi-Hop LoRa Communication for Natural Disaster Management: A Field Test," 2023 International Conference on Information and Communication Technologies for Disaster Management (ICT-DM), Cosenza, Italy, 2023, pp. 1-4, doi: 10.1109/ICT-DM58371.2023.10286960.
- [10] N. G. Agudelo C., J. C. A. Tovar and Y. A. R. Parra, "Alternative Communication System for Emergency Situations," 2023 11th International Japan-Africa Conference on Electronics, Communications, and Computations (JAC-ECC), Alexandria, Egypt, 2023, pp. 115-120, doi: 10.1109/JAC-ECC61002.2023.10479659.
- [11] B. Carneiro, D. M. Batista, R. H. Junior and K. Ullah, "A Crash Response System using LoRa-based V2X Communications," NOMS 2023-2023 IEEE/IFIP Network Operations and Management Symposium, Miami, FL, USA, 2023, pp. 1-5, doi: 10.1109/NOMS56928.2023.10154409.
- [12] J. Thaenthong, T. Pimsen and K. Chidchuea, "R-AODV: Reverse Ad-Hoc On-Demand Distance Vector Routing Algorithm With LoRa Mesh Based for Emergency Alert System," 2023 15th International Conference on Information Technology and Electrical Engineering (ICITEE), Chiang Mai, Thailand, 2023, pp. 143-148, doi: 10.1109/ICITEE59582.2023.10317648.
- [13] L. Qiang, M. Ying, B. Xiujun, L. Xiangning, L. Yan and H. Chunlei, "Research on Localization Techniques in Occluded Environments Based on Lora Signals," 2024 IEEE 14th International Conference on Electronics Information and Emergency Communication (ICEIEC), Beijing, China, 2024, pp. 205-208, doi: 10.1109/ICEIEC61773.2024.10561742.
- [14] S. Thangam, M. Gurupriya, O. Kothari, S. N. Sah, K. P. Shah and K. P. Dharmaraj, "Real Time Accident Alert System in Hilly Region using LoRa Technology," 2024 5th International Conference on Smart Electronics and Communication (ICOSEC), Trichy, India, 2024, pp. 700-706, doi: 10.1109/ICOSEC61587.2024.10722473.
- [15] C. Kumari et al., "Advancing Marine Communication and Sustainability: A LoRa Based Ad Hoc Network System for the Fisheries Industry," 2024 6th International Conference on Advancements in Computing (ICAC), Colombo, Sri Lanka, 2024, pp. 19-24, doi: 10.1109/ICAC64487.2024.10851106.
- [16] M. P. Manuel, M. Faied and M. Krishnan, "A LoRa-Based Disaster Management System for Search and Rescue Mission," in IEEE Internet of Things Journal, vol. 11, no. 20, pp. 34024-34034, 15 Oct.15, 2024, doi: 10.1109/JIOT.2024.3435734.
- [17] T. S. R. Gamidi, T. S. Surya, S. Saragadam and J. J. Janet Kumari, "SafeComm: An IOT-based System for enhancing rural child safety using LoRa Technology," 2024 Global Conference on Communications and Information Technologies (GCCIT), BANGALORE, India, 2024, pp. 1-9, doi: 10.1109/GCCIT63234.2024.10862691.
- [18] M. Sommer, A. Sterz, W. Gellert and B. Freisleben, "Demo: Using Smart Smoke Detectors for Emergency Communication," 2024 IEEE 49th Conference on Local Computer Networks (LCN), Normandy, France, 2024, pp. 1-3, doi: 10.1109/LCN60385.2024.10639738.
- [19] D. Schmidt, F. Kuntke, M. Bauer and L. Baumgärtner, "BPoL: A Disruption-Tolerant LoRa Network for Disaster Communication," 2023 IEEE Global Humanitarian Technology Conference (GHTC), Radnor, PA, USA, 2023, pp. 440-447, doi: 10.1109/GHTC56179.2023.10354717.
- [20] S. Wu, X. Zhou and W. Wang, "Throughput Optimization of IoT System in Disaster Scenario Based on Reinforcement Learning," 2023 International Wireless Communications and Mobile Computing (IWCMC), Marrakesh, Morocco, 2023, pp. 427-432, doi: 10.1109/IWCMC58020.2023.10182863.